

Independent Audit Note on “Cobham-Free Full General Automatic Rigidity for Collatz-Invariant Sequences”

Adversarial examination of cross-base kernel transfer,
dyadic thinness, and topological pumping

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23 September 2026

Post-release supplementary audit note

Abstract

This note records a post-release adversarial audit of three structurally delicate steps in the Cobham-free Full General ARC proof. The audit was prompted by a challenge that proposed possible failures in (i) the transfer from base-2 automaticity of a shortcut-Collatz-invariant sequence to base-3 automaticity of its odd subsequence, (ii) the derivation of dyadic nowhere-thickness for the odd-subsequence change set, and (iii) the theorem asserting that an odd-base automatic Boolean support cannot be both infinite and dyadically nowhere thick. A fourth concern was that the Lean theorem might hold only vacuously because its hypotheses describe an empty class.

No counterexample to the published theorem is established in this note, and no erratum or corrigendum is asserted. Source-level tracing confirms that the second step is implemented through an explicit affine pullback of a previously proved local-density theorem, while the third step is not an unproved cross-base heuristic: the formal development contains a dedicated chain from odd-base pumping to dyadic residue permutation, child filling, and contradiction with nowhere-thickness. A proposed ternary Cantor-type support also fails to supply the claimed counterexample in finite residue tests. The vacuity concern is ruled out constructively by explicit nonconstant-at-zero invariant automatic sequences. The first step is formally connected to the lower theorem `arc_three_kernel_odd_finite`; however, because that lower proof predates the present Cobham-removal package, a fresh line-by-line human re-derivation of that lemma remains a sensible priority for continued audit.

Here “independent” means independent of the main packaged proof endpoint: this is an additional adversarial review of the proof architecture, not an external peer review by an independent third party.

1 Status and purpose

The parent paper proves the following rigidity statement for the shortcut Collatz map

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

For every base $B \geq 2$, if $a : \mathbb{N}_0 \rightarrow \alpha$ has finite base- B kernel and satisfies

$$a(T(n)) = a(n) \quad (n \in \mathbb{N}_0),$$

then a is constant on the positive integers. The value at zero may remain independent.

This note does *not* restate the complete proof. Instead, it asks how the argument could fail if the theorem were false. The audit focuses on the three places where two different structures meet:

- (A1) base-2 finite-kernel structure versus a base-3 odd subsequence;
- (A2) shortcut-Collatz dynamics versus 2-adic cylinders;
- (A3) odd-base finite automata versus dyadic nowhere-thickness.

A fourth check asks whether the hypotheses are nonempty or whether the formal theorem could be vacuous.

Audit target	Present finding	Status
A1. Odd subsequence becomes base-3 finite-kernel automatic	The public Cobham-removal wrapper reduces exactly to <code>arc_three_kernel_odd_finite</code> , using only the even branch, odd branch, and finite base-2 kernel. No Collatz-convergence hypothesis appears in the wrapper or in the recorded axiom audit. A fresh standalone human proof of the lower lemma is still desirable.	No confirmed defect; deeper audit retained.
A2. Change set is dyadically nowhere thick	The source explicitly invokes the local-density theorem with gap $H = 2$ on the affine image $x = 2n + 1$ and constructs a dyadic child in the original n -variable.	Source-level route traced.
A3. Odd-base automatic + dyadically nowhere thick implies finite support	The formal chain explicitly proves dyadic residue permutation, child filling, collision with nowhere-thickness, and then the finite-support theorem.	Strongest remaining target for human re-derivation.
A4. Vacuous truth	Explicit invariant finite-kernel sequences satisfying the hypotheses exist, including a family that may take a separate value at zero.	Ruled out constructively.

2 Audit A1: base-3 automaticity of the odd subsequence

Let $a : \mathbb{N}_0 \rightarrow \alpha$ be shortcut-Collatz invariant and define

$$b(n) = a(2n + 1).$$

The parent proof needs

$$\mathcal{K}_2(a) \text{ finite} \implies \mathcal{K}_3(b) \text{ finite}.$$

The adversarial concern is that repeated interaction between coefficients 2 and 3 might generate infinitely many residual states, or might silently use a global statement about Collatz orbits.

2.1 What the active Cobham-removal wrapper actually assumes

The active Stage-2 wrapper first derives the two branch identities

$$a(2n) = a(n), \tag{1}$$

$$a(2n + 1) = a(3n + 2), \tag{2}$$

from shortcut invariance. It then unfolds finite-kernel automaticity to obtain

$$(\mathcal{K}_2(a)).\text{Finite}$$

and calls the lower theorem

$$\text{arc_three_kernel_odd_finite.}$$

The output is precisely

$$(\mathcal{K}_3(n \mapsto a(2n + 1))).\text{Finite.}$$

The resulting wrapper is

$$\text{arc2OddSubseq_threeAutomatic_of_twoAutomatic.}$$

The recorded Lean axiom report for `arc_three_kernel_odd_finite` contains only the standard logical/quotient dependencies

$$\text{propext, Classical.choice, Quot.sound.}$$

Thus the active proof does not register a Collatz-convergence principle, a Cobham principle, or an unproved custom axiom at this step.

2.2 What this does and does not settle

This source trace defeats one specific criticism: the Stage-2 proof is not merely assuming that Collatz orbits “eventually fall into finitely many patterns.” The finite base-3 kernel is supplied by a concrete lower Lean theorem whose inputs are the branch identities and finite base-2 kernel.

However, formal dependency tracing is not the same as a new human proof. The lower theorem belongs to the earlier ARC2 kernel-transfer development and is reused here. Accordingly, the strongest responsible conclusion of this audit is:

No defect has been identified in the active transfer step, but a fresh line-by-line mathematical reconstruction of `arc_three_kernel_odd_finite`, independent of its original development, remains a high-value audit task.

This is a retained audit item, not a correction notice.

3 Audit A2: dyadic thinness of the odd-subsequence change set

Define the adjacent-change set

$$\mathcal{D}_b = \{n \in \mathbb{N}_0 : b(n) \neq b(n + 1)\}.$$

The concern is that shortcut Collatz dynamics involves multiplication by 3, whereas the thinness statement is expressed in dyadic residue classes. A purely heuristic appeal to “compatibility” between these structures would indeed be insufficient.

3.1 The actual affine pullback

The active Stage-1 proof does not assume such compatibility. It explicitly applies the previously proved local-density theorem

$$\text{arc5_naturalInvariantBlock_in_dyadicCylinder}$$

with gap parameter $H = 2$ to the odd affine image. Starting from a prescribed parent class

$$n \equiv r \pmod{2^m},$$

the map

$$x = 2n + 1$$

sends it into

$$x \equiv 2r + 1 \pmod{2^{m+1}}.$$

Inside a deeper dyadic child, the local-density theorem supplies

$$a(x) = a(x + 2).$$

After substituting $x = 2n + 1$, this is exactly

$$a(2n + 1) = a(2n + 3),$$

hence

$$b(n) = b(n + 1).$$

The source then pulls the child cylinder back to the n -variable and proves that it remains inside the original parent cylinder. Therefore that child is disjoint from $\mathcal{D}_b$.

The public endpoint is

$$\text{arc2OddSubseq_changeSet_nowhereThick},$$

packaged as

$$\text{arc2CobhamRemoval_stage1_thinness}.$$

3.2 Audit conclusion for A2

The objection “base 2 and multiplication by 3 are transversal, therefore the child need not exist” does not identify a missing inference in the active proof. The existence of the child is precisely what the earlier local-density machinery proves, and the affine conversion is carried out explicitly in the Stage-1 source.

The natural next adversarial question is therefore not whether the two bases are different, but whether the local-density theorem itself has been modeled correctly and proved from the stated shortcut invariance. That theorem was already developed independently of automaticity in the ARC5 local-density chain, so it can be audited as a separate arithmetic lemma family.

4 Audit A3: odd-base automatic support versus dyadic nowhere-thickness

This is the most structurally delicate point. The relevant theorem states, for odd $q \geq 3$, that if a Boolean sequence has finite base- q kernel and its support is dyadically nowhere thick, then its support is finite.

At first sight this is surprising: the automaton reads base- q digits, while thinness is measured using powers of 2. The audit therefore asks whether the proof actually bridges these incompatible scales, or merely assumes that it can.

4.1 The formal bridge is explicit

The General ARC chain contains the following dedicated stages.

1. **Dyadic residue permutation.** For an odd base, the normalized doubled-loop pumped family defines a map on residues modulo 2^M . The theorem `arcGeneral_oddPumpResidueMap_injective` proves injectivity; finiteness of the common domain and codomain yields surjectivity; hence every residue modulo 2^M is hit.
2. **Dyadic child filling.** The theorem `arcGeneral_evenLoopPumpedFamily_hits_every_child_of_exact_two_factor` lifts residue surjectivity into the assertion that the pumped family hits every deeper dyadic child of the relevant parent cylinder.
3. **Collision with nowhere-thickness.** The theorem `arcGeneral_child_filling_not_nowhereThick` assumes a pumped family contained in the support, selects the child that thinness claims is empty, and then uses child filling to produce a support point in that same child, a contradiction.
4. **Infinite support gives a long pumpable word.** Stage 9 extracts a sufficiently long canonical support word from any infinite automatic support and feeds it into the Stage-8 collision theorem.
5. **Finite-kernel handoff.** Stage 10 converts finite kernel to an MSD DFAO and applies the preceding topological-pumping theorem. Its public endpoint is `arcGeneral_kernelAutomatic_nowhereThick_support_finite`.

Thus the base mismatch is not ignored; it is the central arithmetic object of Stages 6–10.

4.2 Adversarial Cantor-style test

A natural proposed counterexample is the base-3 digit-restricted set

$$C = \left\{ \sum_{i \geq 0} \varepsilon_i 3^i : \varepsilon_i \in \{0, 1\}, \text{ finitely many } \varepsilon_i \neq 0 \right\}.$$

This set is base-3 automatic and infinite. The proposed attack asks whether it might also be dyadically nowhere thick.

A finite residue check gives the opposite indication. Let

$$C_L = \left\{ \sum_{i=0}^{L-1} \varepsilon_i 3^i : \varepsilon_i \in \{0, 1\} \right\}.$$

For each modulus 2^m in the following range, a finite L already makes C_L hit every residue class modulo 2^m :

m	1	2	3	4	5	6	7	8	9
smallest tested L	1	3	4	5	7	8	9	11	12

For $m = 10$, the smallest tested value is also $L = 12$. These are exact finite computations of residue coverage, not random sampling. They do not constitute a proof for all m , but they show that this particular Cantor-style candidate is not supported by the suggested finite test. In particular, an earlier finite search that reported failure to find an empty child in the parent $0 \pmod{2}$ points against, rather than for, dyadic nowhere-thickness.

4.3 Audit conclusion for A3

No counterexample has been identified. The formal proof contains an explicit mixed-base mechanism whose purpose is to turn odd-base pumping into complete coverage of dyadic children. Nevertheless, because this mechanism is both novel within the ARC development and mathematically nontrivial, A3 remains the highest-priority target for a completely independent human proof written without following the Lean file order.

5 Audit A4: vacuous-truth concern

The suggestion that the Lean theorem might be true only because no sequence satisfies its hypotheses can be rejected constructively.

Fix arbitrary $x, y \in \alpha$ and define

$$a(0) = x, \quad a(n) = y \quad (n > 0).$$

Then $T(0) = 0$, while $T(n) > 0$ for every $n > 0$. Hence

$$a(T(n)) = a(n)$$

for all $n \geq 0$. Moreover, in every base $B \geq 2$, every kernel residual of a is either the original sequence a or the constant sequence $n \mapsto y$. Thus its base- B kernel has at most two states.

If $x \neq y$, this sequence is not globally constant on $\mathbb{N}_0$, but it is constant on the positive integers exactly as the theorem predicts. Therefore the hypothesis class is manifestly nonempty, and the theorem is not a vacuous implication caused by an inconsistent or empty premise.

6 What a genuine counterexample would have to provide

To refute the Full General ARC theorem, it is not enough to propose a coloring that is constant on Collatz orbits, or a sparse automatic set, or a coloring that respects only the even branch. A genuine counterexample must explicitly exhibit some $B \geq 2$ and a sequence $a : \mathbb{N}_0 \rightarrow \alpha$ satisfying all three conditions simultaneously:

- (i) the full base- B kernel $\mathcal{K}_B(a)$ is finite;
- (ii) $a(T(n)) = a(n)$ for every $n \geq 0$;
- (iii) there exist positive m, n with $a(m) \neq a(n)$.

For example, assigning colors to hypothetical nonstandard Collatz components might satisfy (ii), but it says nothing by itself about the finite-kernel condition (i). Likewise, extending a coloring by $a(2^k q) = a(q)$ enforces the even relation but does not automatically enforce the odd relation

$$a(2n + 1) = a(3n + 2).$$

Any claimed counterexample must survive these exact checks.

7 Lean 4: what formal verification does and does not certify

The completed development reported successful project builds and no `sorryAx` at the final audited endpoints. The final theorem also no longer carries an explicit `ARCCobhamPrinciple` argument or a `[Finitealpha]` hypothesis. This is meaningful evidence that the packaged theorem follows from the definitions and imported lemmas accepted by Lean.

Formal verification does not, however, replace the following external tasks:

- checking that the formal definitions express the intended mathematical notions;
- checking the exposition of imported lower lemmas against an independent paper proof;
- checking novelty and relation to the literature;
- searching for an unintended theorem that is true for a different reason than the intended argument.

The present note is directed at the first two items. Its role is therefore supplementary: it records where the proof was attacked, what the source-level trace showed, and which points still merit independent mathematical reconstruction.

8 Current audit judgment and next steps

The adversarial review has not produced a confirmed error in the published Full General ARC theorem. It has instead isolated three high-value audit points. The present status is:

1. **A1: retained for deeper audit.** Re-derive `arc_three_kernel_odd_finite` from its kernel definitions on paper, without invoking the packaged theorem.
2. **A2: source-level path confirmed.** The affine dyadic pullback is explicit; a future audit should target the underlying local-density theorem itself rather than the mere fact that bases 2 and 3 differ.
3. **A3: highest priority.** Reconstruct the residue-permutation and child-filling argument independently, and actively search for an infinite odd-base automatic support that could evade the proof.
4. **A4: closed.** Vacuous truth is ruled out by explicit witnesses.

Accordingly, this document is an *Independent Audit Note*, not an erratum, corrigendum, or replacement of the parent paper. If a genuine defect is later identified, it should be recorded separately and transparently as a correction.

Reproducibility note

The finite Cantor-style residue experiment used the exact sets

$$C_L \bmod 2^m$$

formed by all 2^L choices of digits $\varepsilon_i \in \{0, 1\}$, with set compression after each added power of 3. The reported smallest L values are therefore exhaustive for the stated moduli. They are included only as an adversarial sanity check and are not used in the Lean proof.

Acknowledgment of AI assistance

Generative AI systems were used to formulate adversarial objections, inspect proof architecture, assist with Lean-source tracing, run finite computational checks, and prepare this note. An AI-generated objection is not treated as a mathematical result unless its claim is independently verified. Responsibility for the released mathematical statements and the interpretation of the formal artifacts remains with the human author.

References

- [1] Muteki, *Cobham-Free Full General Automatic Rigidity for Collatz-Invariant Sequences*, Full General ARC release v1.0.0, 23 September 2026.
- [2] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [3] A. Cobham, On the base-dependence of sets of numbers recognizable by finite automata, *Mathematical Systems Theory* **3** (1969), 186–192.